Coupon Collector's Problem

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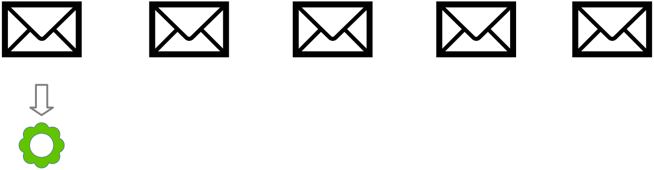
Our memories (2005) – Hello Kitty Magnets

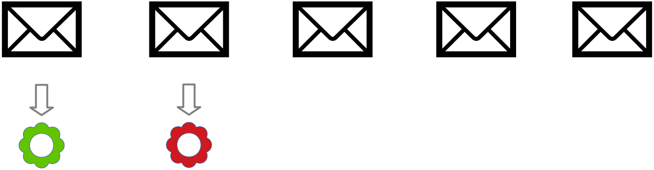
https://a.rimg.com.tw/s1/7/f7/89/10061029175177_650.jpg

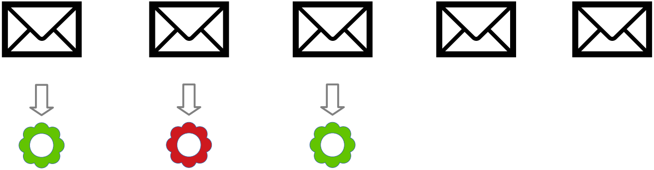
Outline

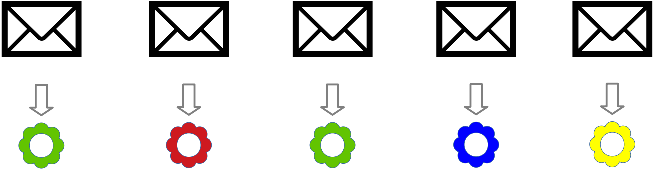
1. Coupon Collector's Problem
2. Idea from Geometric Distributions
3. Conditional Expectations

Coupon Collector's Problem









Coupon Collector's Problem

- We want to collect all of the n types of the coupons (i.e., Hello Kitty magnets).
- Have you thought about how much you should roughly pay for them?

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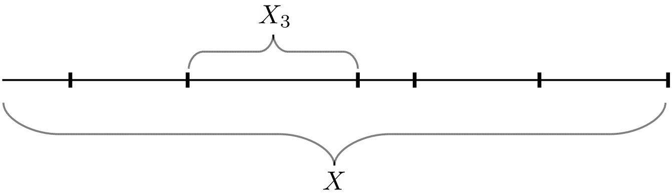
- Have you thought about how much you should roughly pay for them?
  - You need to pay NT\$ 77 to get one coupon.

Coupon Collector's Problem (stage by stage)



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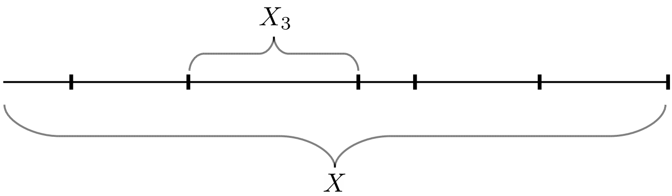

- X: the number of bags bought until every type of coupon is obtained.

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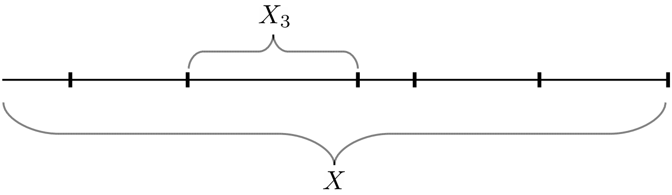

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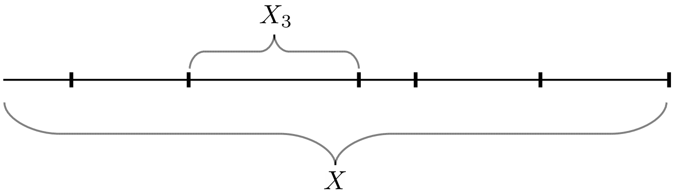

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  - $X = \sum_{i=1}^{n} X_i$.

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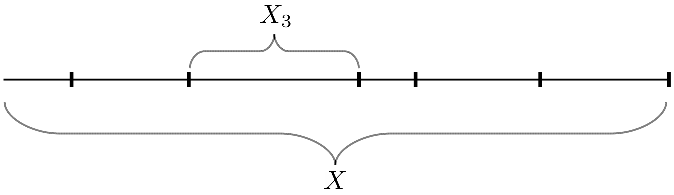

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  - $X = \sum_{i=1}^{n} X_i$.
  - X_i: Geometric random variables

Geometric Distribution

Consider the scenario:

- flip a coin until it lands on a head

- What's the distribution of the number of flips?

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Geometric random variable

For a geometric random variable X with parameter p, we have

$$\Pr[X = n] = (1-p)^{n-1}p.$$

for $n = 1, 2, \ldots$

Properties of a geometric random variable

Memorylessness

Let X be a geometric random variable X with parameter $p > 0$, we have

$$\Pr[X = n + k \mid X > k] = \Pr[X = n]$$

for any $n, k > 0$.

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Expected Value

Let X be a geometric random variable X with parameter $p > 0$, we have

$$\mathbb{E}[X] = \frac{1}{p}.$$

Memoryless (Proof)

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$$\Pr[X = n + k \mid X > k] = \frac{\Pr\big[(X = n + k) \cap (X > k)\big]}{\Pr[X > k]}$$

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&= (1-p)^{n-1}p \\
&= \Pr[X = n].
\end{aligned}
$$

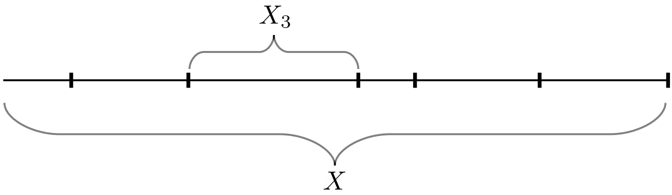
The mean of a geometric r.v. $X(p)$

$$\begin{aligned}\mathbb{E}[X] &= \sum_{j=1}^{\infty} j \Pr[X=j] \\ &= \sum_{j=1}^{\infty}\sum_{i=1}^{j} \Pr[X=j] \\ &= \sum_{i=1}^{\infty}\sum_{j\geq i} \Pr[X=j] \\ &= \sum_{i=1}^{\infty} \Pr[X\geq i].\end{aligned}$$

$$\begin{aligned}\mathbb{E}[X] &= \sum_{i=1}^{\infty}(1-p)^{i-1} \\ &= \frac{1}{1-(1-p)} \\ &= \frac{1}{p}.\end{aligned}$$

$$\Pr[X\geq i] = \sum_{k=i}^{\infty}(1-p)^{k-1}p = (1-p)^{i-1}.$$

Back to the Coupon Collector's Problem



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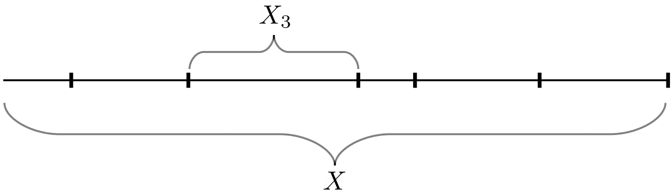

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- Since each X_i is a geometric random variable, we gave

$$\mathbf{E}[X_i] = \frac{1}{p_i} = \frac{n}{n-i+1}.$$

The last steps

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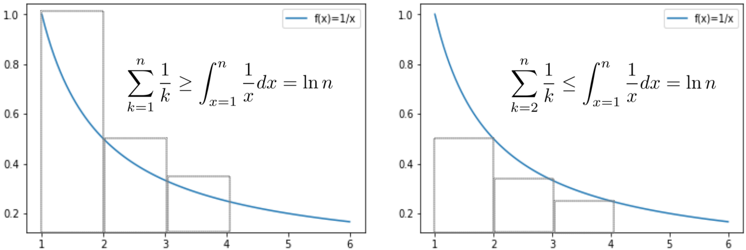
The last steps

$$\begin{aligned}\mathbb{E}[X] &= \mathbb{E}\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} \mathbb{E}[X_i] = \sum_{i=1}^{n} \frac{n}{n-i+1} = n \cdot \sum_{i=1}^{n} \frac{1}{i} \\ &= n(\ln n + \Theta(1)).\end{aligned}$$



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- How likely is this to happen?
  - $\Pr[X \geq 2R] \leq \frac{1}{2}$ (Markov Inequality).

- Let X be a random variable that assumes only non-negative values. Then, for all $a > 0$,

$$\Pr[X \geq a] \leq \frac{\mathbf{E}[X]}{a}.$$

Markov Inequality

Andrei Andreyevich Markov (Wikipedia)
1856–1922

This can be very likely to happen actually...

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  - Getting $2R$ coupons costs you $\approx$ 23,700.
- How likely is this to happen?
  - $\Pr[X \geq 2R] \leq \frac{\pi^2}{6 \ln^2 34} < 0.1323$ (Chebyshev's Inequality).

<u>Theorem</u> [Chebyshev's Inequality]. For any $a > 0$,

$$\Pr[|X - \mathbf{E}[X]| \geq a] \leq \frac{\mathbf{Var}[X]}{a^2}.$$


Learn the coupon collector's problem well!
Pay me NT$ 30,000 and you get them all...

Variance of a geometric random variable

Variance

Let X be a geometric random variable X with parameter $p > 0$, we have

$$\mathbf{Var}[X] = \frac{1-p}{p^2}.$$

Let $X_1, X_2, \ldots, X_n$ be n independent geometric random variables with parameters $p_1, p_2, \ldots, p_n > 0$ respectively. Let $X = \sum_{i=1}^{n} X_i$, then we have

$$\mathbf{Var}[X] = \sum_{i=1}^{n} \frac{1-p_i}{p_i^2} < \sum_{i=1}^{n} \frac{1}{p_i^2}.$$

$$\mathbf{Var}[X] = \sum_{i=1}^{n} \mathbf{Var}[X_i] < \sum_{i=1}^{n} \left(\frac{1}{p^2}\right) = \sum_{i=1}^{n} \left(\frac{n}{n-i+1}\right)^2$$
$$= n^2 \sum_{i=1}^{n} \left(\frac{1}{i}\right)^2$$

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&= n^2 \sum_{i=1}^{n} \left(\frac{1}{i}\right)^2 \\
&\leq \frac{\pi^2 n^2}{6}. \quad \text{(Leonhard Euler (1740))}
\end{aligned}$$

Remark

- Fix one coupon type. The probability that it is not collected after choosing $R = 2n \ln n$ draws:

$$\left(1 - \frac{1}{n}\right)^{2n \ln n} \leq e^{-2 \ln n} = \frac{1}{n^2}.$$

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- Thus, by the union bound,

$$\Pr[\text{some coupon type is missing}] \leq n \cdot \frac{1}{n^2} = \frac{1}{n}.$$

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- Thus, by the union bound,

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- Hence, after $2n \ln n$ draws, all coupons are collected with probability at least $1 - \frac{1}{n}$.

Exercise

What is $\mathbb{E}[X]$ when you have already collected 10 types of coupons?

Outline

1. Coupon Collector's Problem
2. Idea from Geometric Distributions
3. Conditional Expectations

Conditional Expectation

Definition

$$\mathbb{E}[Y \mid Z = z] = \sum_{y} y \Pr[Y = y \mid Z = z].$$

- Example of two dice.
  - X_1: the number showing on the first die
  - X_2: the number showing on the second die
  - $X = X_1 + X_2$

$$\mathbb{E}[X \mid X_1 = 2] = \sum_{x=3}^{8} x \cdot \frac{1}{6} = \frac{11}{2}.$$

Conditional Expectation (contd.)

Lemma (Law of Total Expectation)

For any random variables X and Y,

$$\mathbb{E}[X] = \sum_{y} \Pr[Y = y]\, \mathbb{E}[X \mid Y = y].$$

$$\sum_{y} \Pr[Y = y] \cdot \mathbb{E}[X \mid Y = y] =$$

Conditional Expectation (contd.)

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&= \sum_x \sum_y x \Pr[X = x \mid Y = y] \cdot \Pr[Y = y] \\
&= \sum_x \sum_y x \Pr[\{X = x\} \cap \{Y = y\}]
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Conditional Expectation (contd.)

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&= \sum_x \sum_y x \Pr[X = x \mid Y = y] \cdot \Pr[Y = y] \\
&= \sum_x \sum_y x \Pr[\{X = x\} \cap \{Y = y\}] \\
&= \sum_x x \Pr[X = x] = \mathbb{E}[X].
\end{aligned}$$

Remark

By marginalization of the joint distribution,

$$\sum_{y} \Pr[\{X = x\} \cap \{Y = y\}] = \Pr[X = x].$$

Conditional Expectation

Lemma

For any finite collection of discrete random variables $X_1, X_2, \ldots, X_n$ with finite expectations and for any random variable Y,

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i \,\middle|\, Y = y\right] = \sum_{i=1}^{n} \mathbb{E}[X_i \mid Y = y].$$

Conditional Expectation (contd.)

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- It may seem a bit strange to you to see:
  $\mathbb{E}[Y \mid Z]$: regarded as a random variable $f(Z)$.
  - It takes on the value $\mathbb{E}[Y \mid Z = z]$ when $Z = z$.
- In the previous example,

$$\mathbb{E}[X \mid X_1] = \sum_x x \cdot \Pr[X = x \mid X_1] = \sum_{x=X_1+1}^{X_1+6} x \cdot \frac{1}{6} = X_1 + \frac{7}{2}.$$

- So it makes sense that

$$\mathbb{E}[\mathbb{E}[X \mid X_1]] = \mathbb{E}\left[X_1 + \frac{7}{2}\right] = \frac{7}{2} + \frac{7}{2} = 7.$$

Conditional Expectation (contd.)

Theorem

$\mathbb{E}[Y] = \mathbb{E}[\mathbb{E}[Y \mid Z]]$.

- *Proof.*
$$\mathbb{E}[\mathbb{E}[Y \mid Z]] = \sum_{z} \mathbb{E}[Y \mid Z = z] \cdot \Pr[Z = z] = \mathbb{E}[Y].$$

Conditional Expectation (contd.)

Theorem

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- *Proof.*
$$\mathbb{E}[\mathbb{E}[Y \mid Z]] = \sum_z \mathbb{E}[Y \mid Z = z] \cdot \Pr[Z = z] = \mathbb{E}[Y].$$
- The last equality follows from the law of total expectation.

Discussions